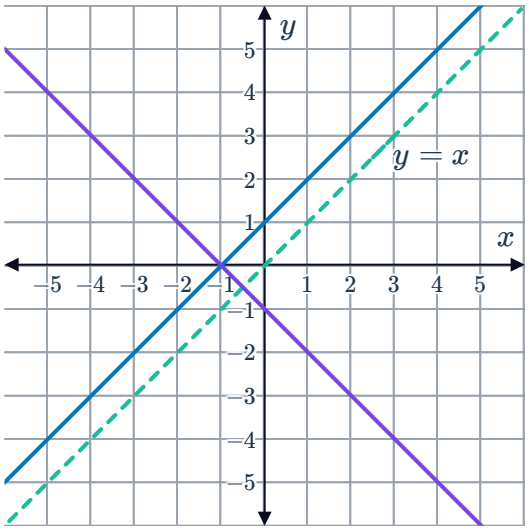
Inverse relations and functions

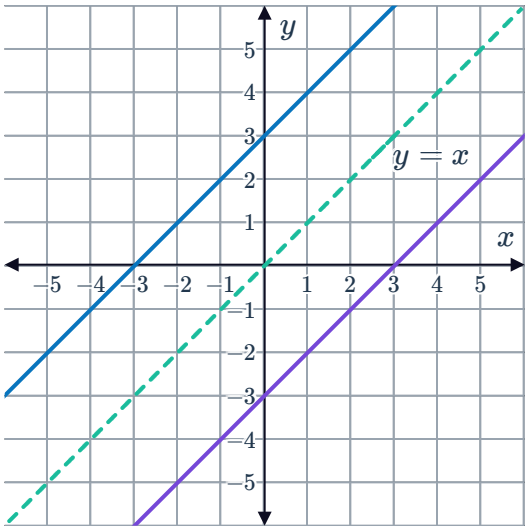
1 The graphs of a function f and its inverse f^{-1} are symmetric about which line?

2 In each graph below two functions and the line $y = x$ are drawn. For each graph, state whether the functions are inverse functions of each other.

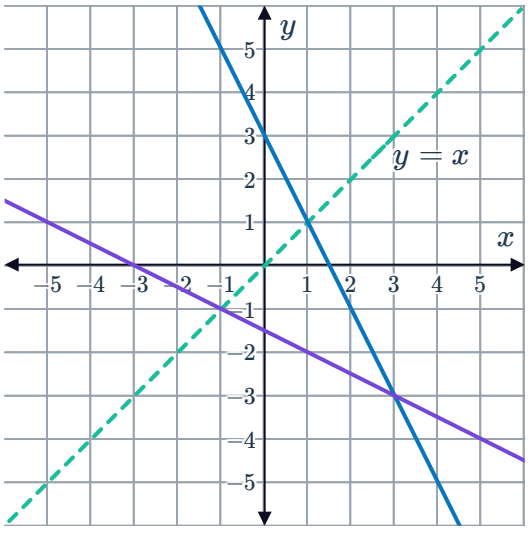
a



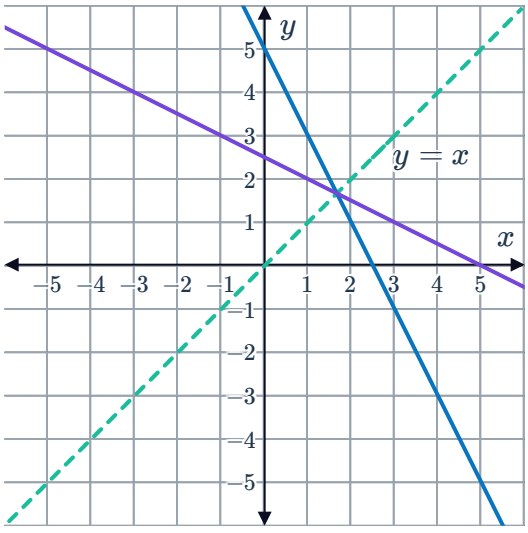
b



c

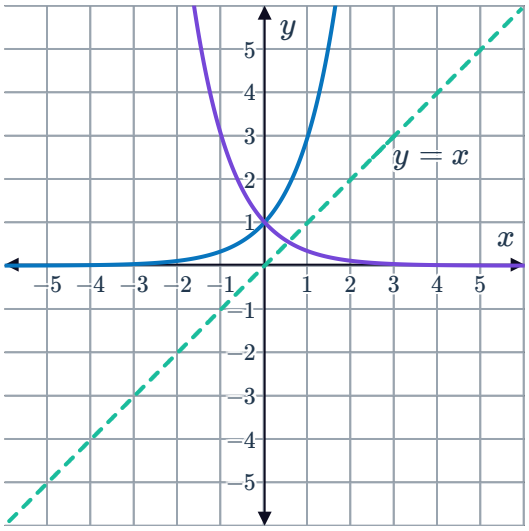


d

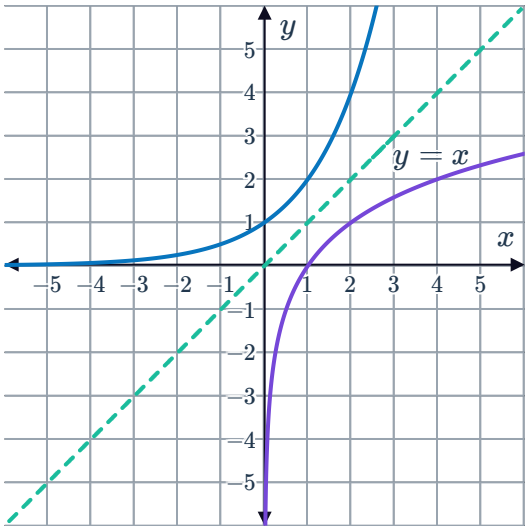


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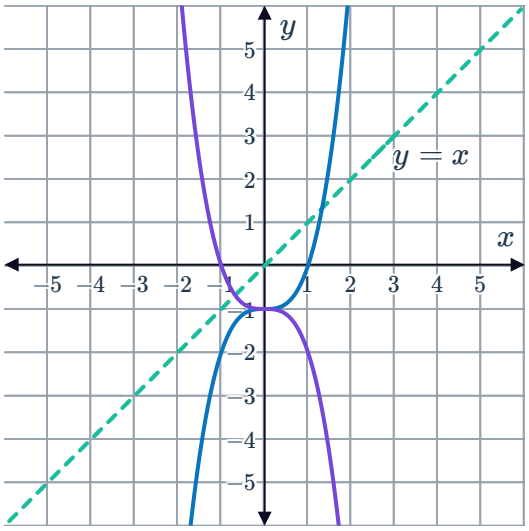
e



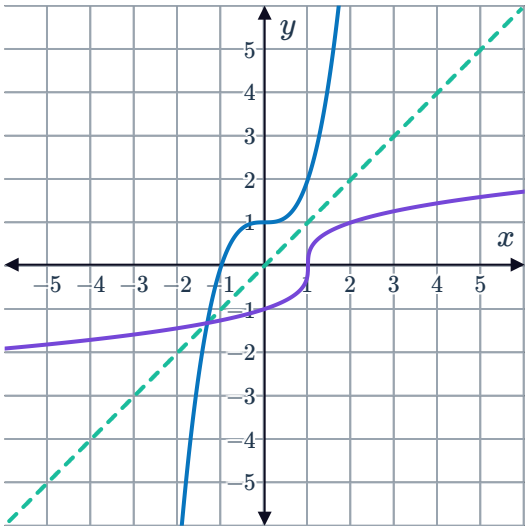
f



g

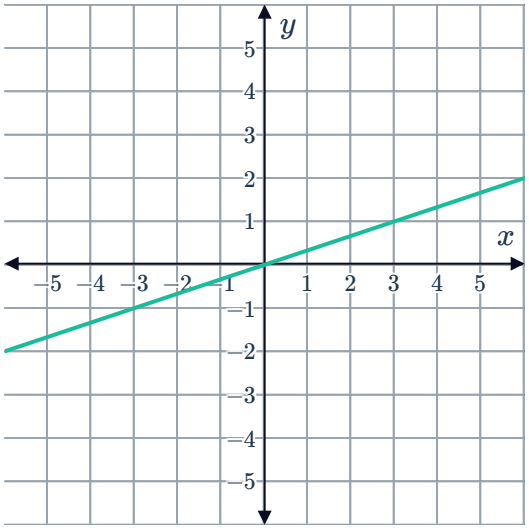


h

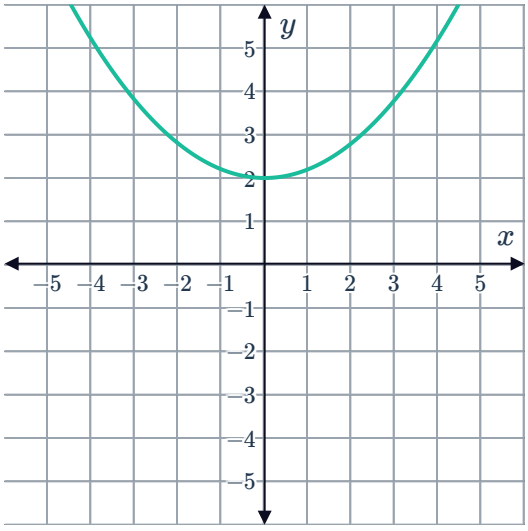


3 State whether the following functions have an inverse function:

a

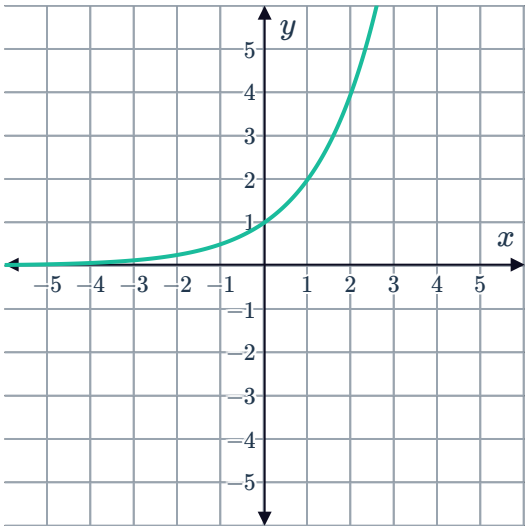


b

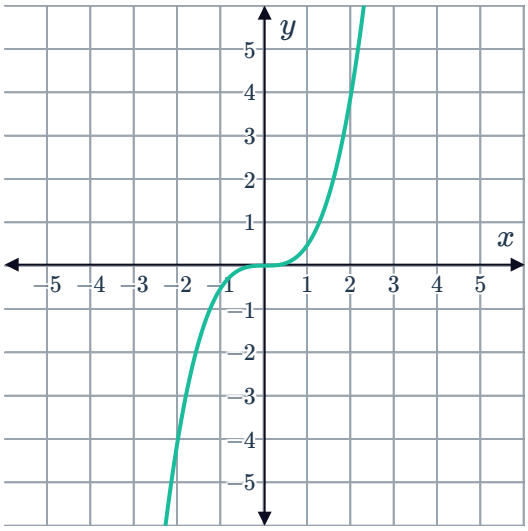


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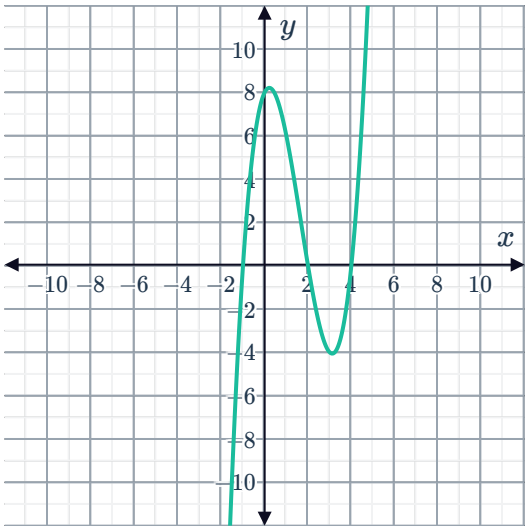
c



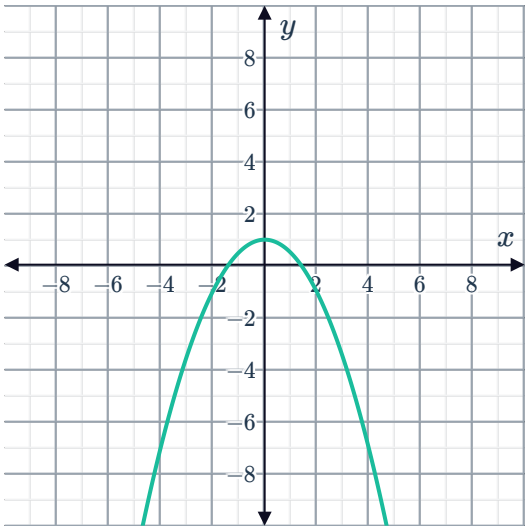
d



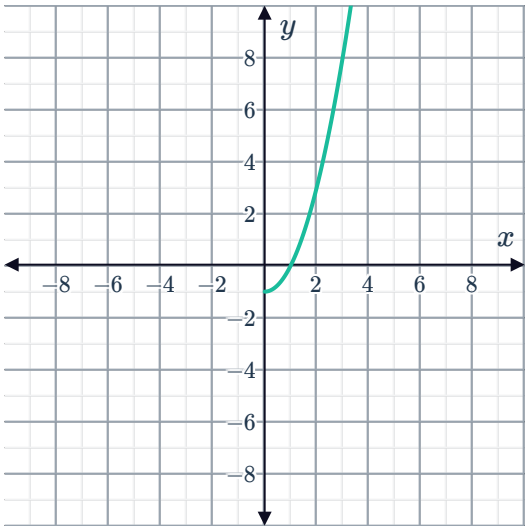
e



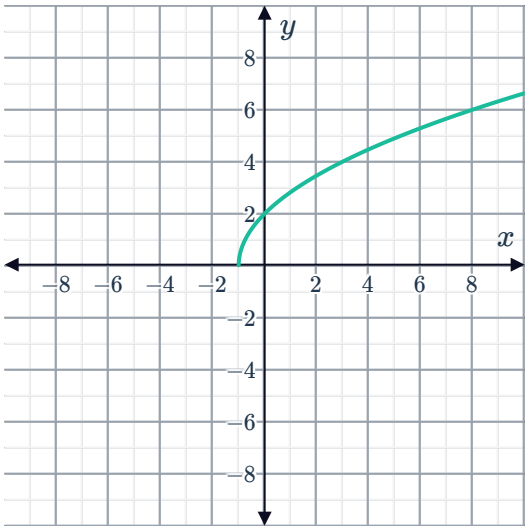
f



g

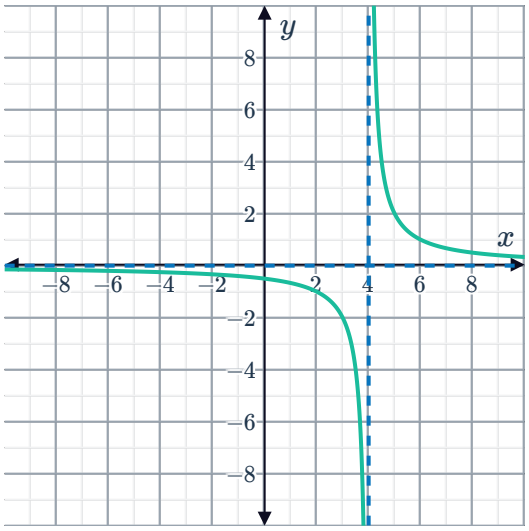


h

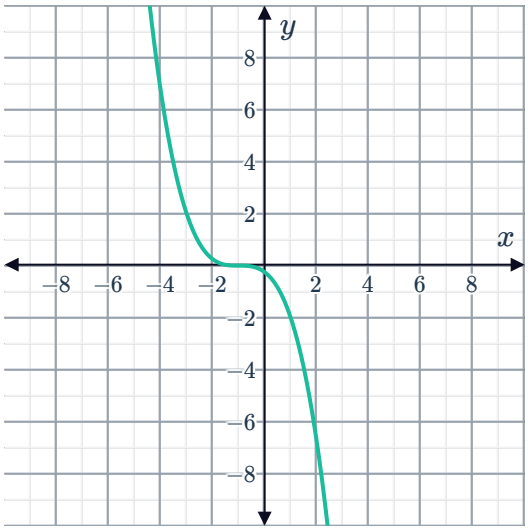


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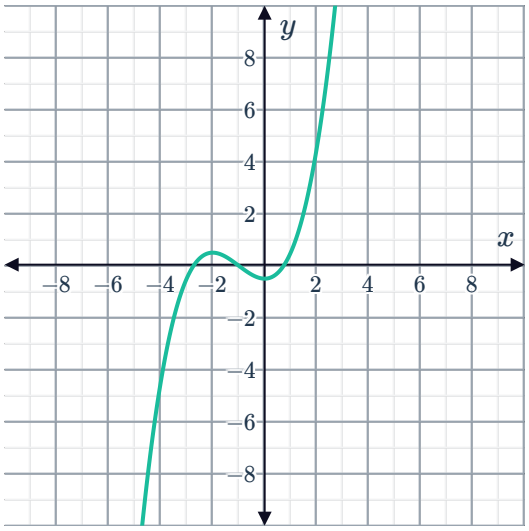
i



j

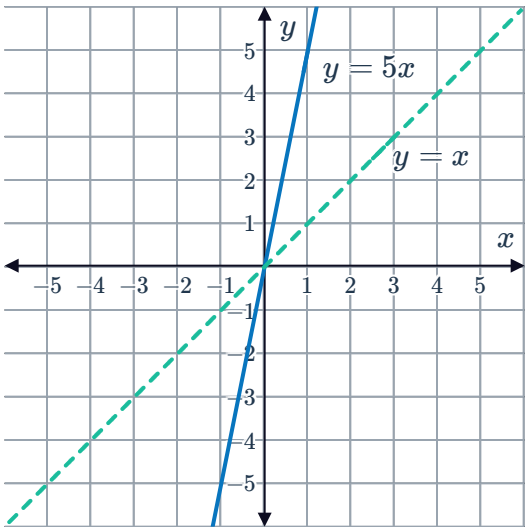


k

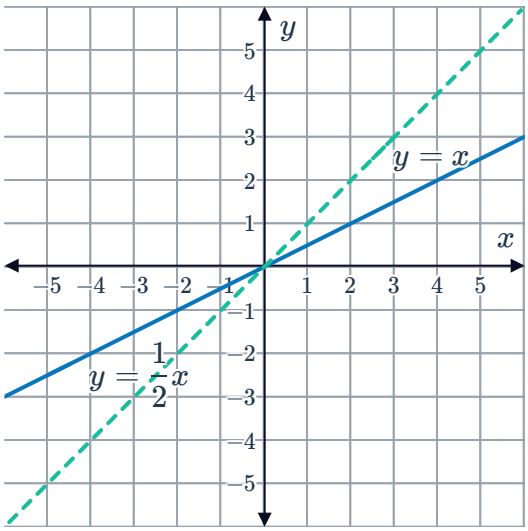


4 Sketch the graph of the inverse of the following functions:

a

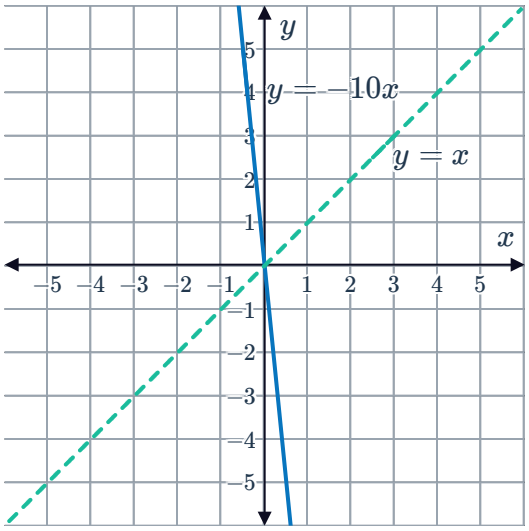


b

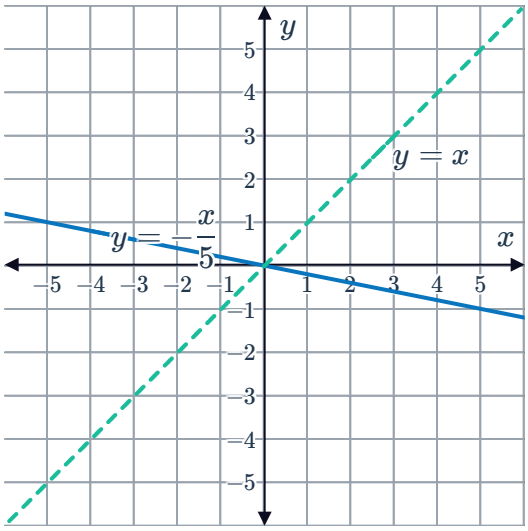


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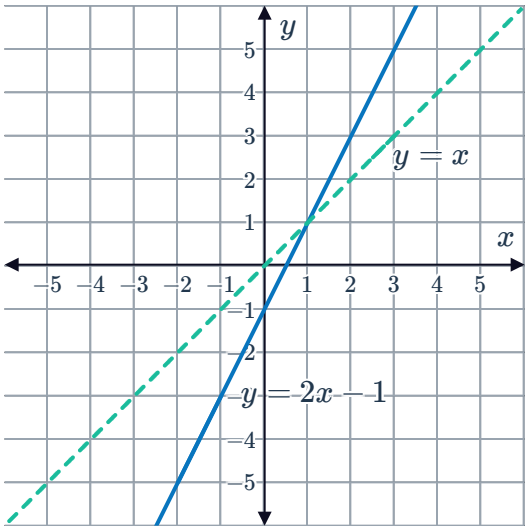
c



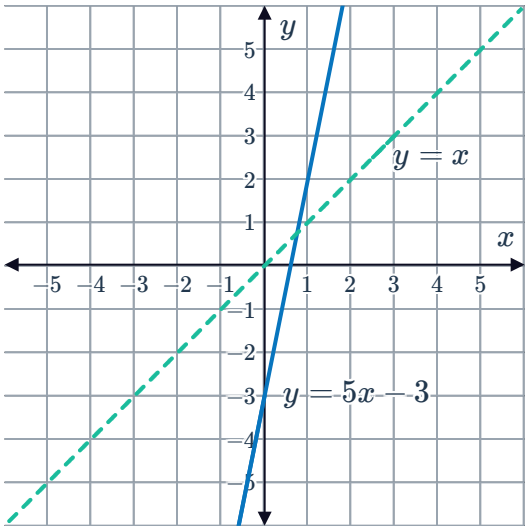
d



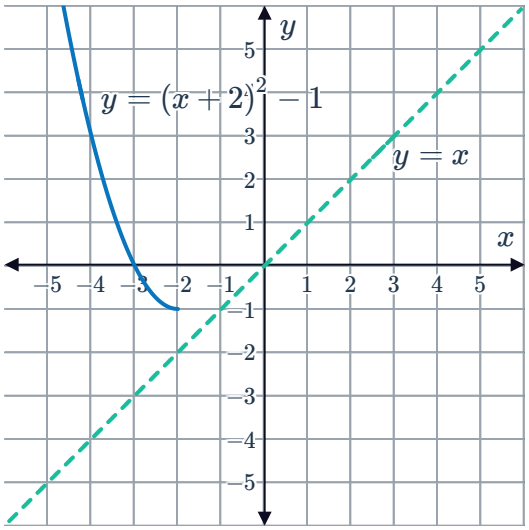
e



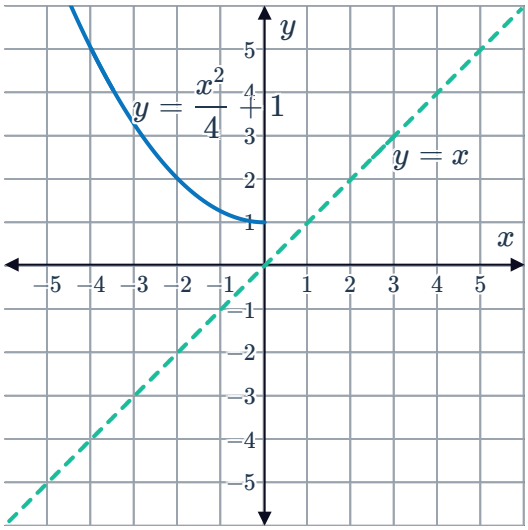
f



g

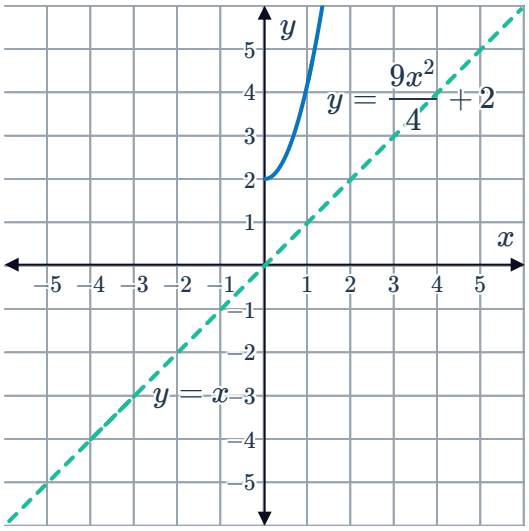


h

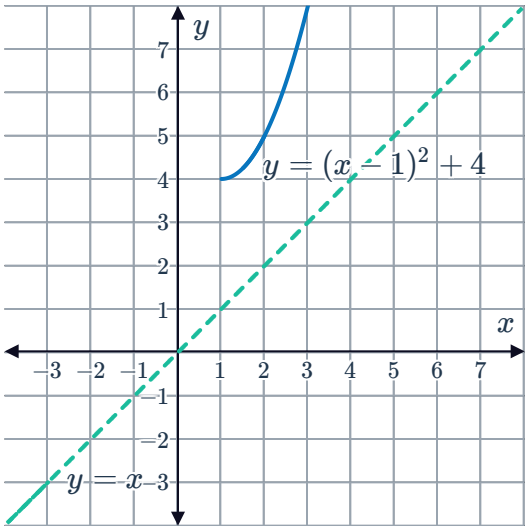


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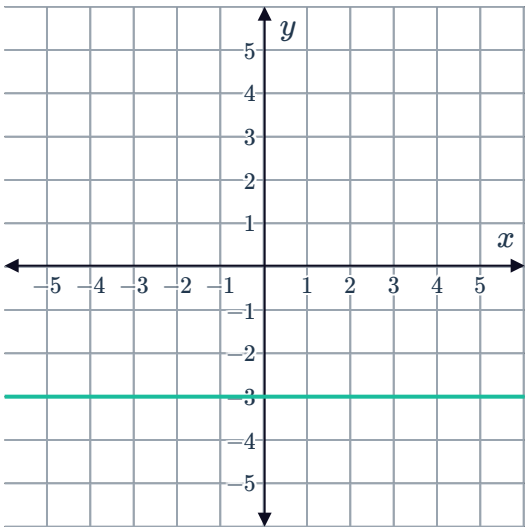
i



j



5 Consider the given graph:



a Does this graph represent a function or just a relation?

b If we try to invert this function, do we get a function or just a relation?

6 For each of the following pairs of functions:

i Find $f(g(x))$.

ii Find $g(f(x))$.

iii Are $f(x)$ and $g(x)$ inverses?

a $f(x) = -\frac{7}{2}x$ and $g(x) = -\frac{2}{7}x$

b $f(x) = 3x - 6$ and $g(x) = \frac{1}{3}x - 2$

c $f(x) = 3x + 12$ and $g(x) = \frac{1}{3}x - 4$

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7 For each of the following functions, find an expression for the inverse function $y = f^{-1}(x)$:

a $f(x) = 5x$

b $f(x) = 7x - 8$

c $f(x) = \dfrac{4x}{3} - 5$

d $f(x) = 6x^3$

e $f(x) = 1 - \dfrac{x}{8}$

f $f(x) = \dfrac{x - 7}{6}$

g $f(x) = \sqrt[3]{x - 2}$

h $f(x) = (x - 4)^3 + 3$

i $f(x) = \dfrac{5x}{x + 2}$

j $f(x) = \dfrac{8}{x - 9} + 5$

k $f(x) = \dfrac{8}{5x + 3}$

l $f(x) = \dfrac{7}{(x + 8)^3} - 3$

m $f(x) = 4(9x - 2)^3 - 7$

Domain and range

8 The graph of a function f is defined by the single point $(-3, 6)$.

a State the domain of the function f.

b State the range of the function f.

c The point is reflected about the line $y = x$ to form the graph of the function f^{-1}. State the coordinates of the single point that defines f^{-1}.

d State the domain of the function f^{-1}.

e State the range of the function f^{-1}.

f What do you notice about the domain and range of f and f^{-1}?

9 A one-to-one function f has a domain of $\{-7, 2\}$ and a range of $\{6, -5\}$. Find the:

a Domain of f^{-1}.

b Range of f^{-1}.

10 Consider the function f given by the set of ordered pairs $\{(5, 9), (7, -4), (8, 2)\}$.

a State the domain of f.

b State the range of f.

c Find the inverse f^{-1}.

d State the domain of f^{-1}.

e State the range of f^{-1}.

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11 Consider the function $f(x) = ax + b$ defined over $[-5, 5]$ where a and b are non-zero constants.

a Using the fact that $f(1) = 2$ and $f^{-1}(1) = 3$, find the values of a and b.

b State the domain of f^{-1} in interval notation.

c State the range of f^{-1} in interval notation.

12 For each of the following functions, find:

i $f^{-1}(x)$ **ii** The domain of $f^{-1}(x)$.

iii The range of $f^{-1}(x)$.

a $f(x) = 5x + 15$ defined over all real values of x

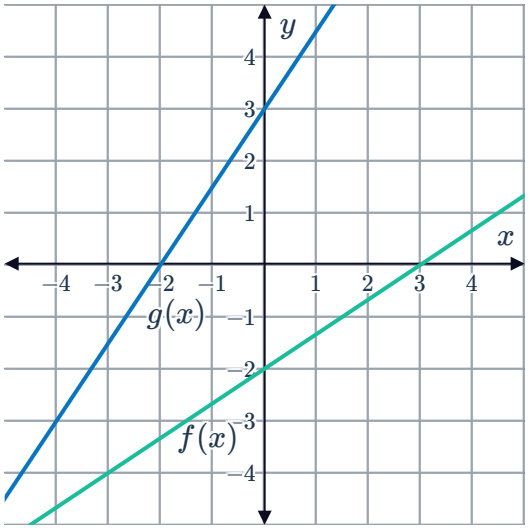
b $f(x) = 4 - 3x$ defined over all real values of x

c $f(x) = 8x - 9$ defined over $[-4, 2]$

d $f(x) = x^2$ defined over $[0, \infty)$

e $f(x) = \sqrt{16 - x^2}$ defined over $[0, 4]$

13 Consider the graphs of $f(x)$ and $g(x)$ shown:



a State the equation of:

i $f(x)$ **ii** $g(x)$

b Evaluate:

i $f(g(x))$ **ii** $g(f(x))$

c State whether the following statements are true:

i $g(x)$ is an inverse of $f(x)$.

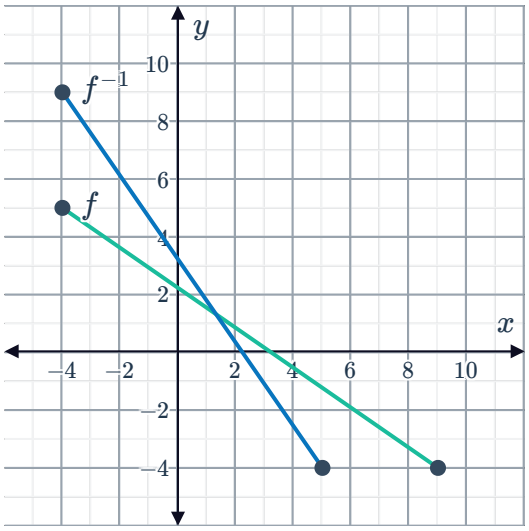
ii $f(x)$ is an inverse of $g(x)$.

iii $f(g(x))$ has slope $\frac{2}{3}$.

iv $g(f(x))$ has slope 1.

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14 The graph of f is represented by the line segment joining $(-4, 5)$ and $(9, -4)$ while f^{-1} is represented by the line segment joining $(5, -4)$ and $(-4, 9)$.



a State the domain of f.

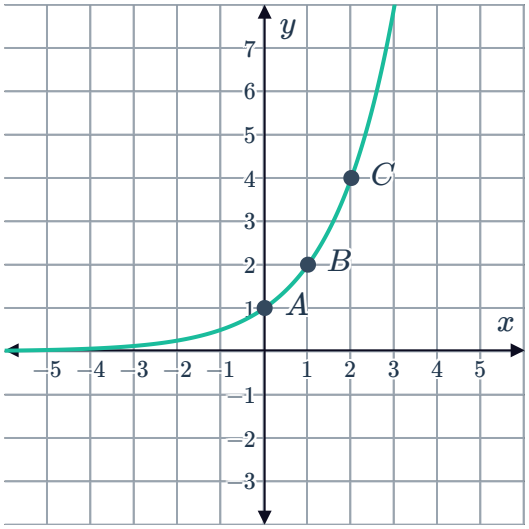
b State the range of f.

c State the domain of f^{-1}.

d State the range of f^{-1}.

e True or False: The range of f is the same as the range of f^{-1}.

15 Consider the graph of the function $f(x)$:



a State the coordinates of the points labeled:

i A ii B iii C

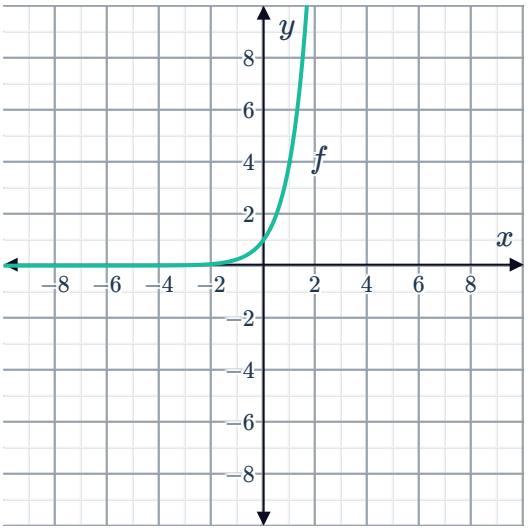
b State the corresponding ordered pairs for f^{-1}, the inverse of f:

i A' ii B' iii C'

c Plot the three inverse points on a coordinate plane.

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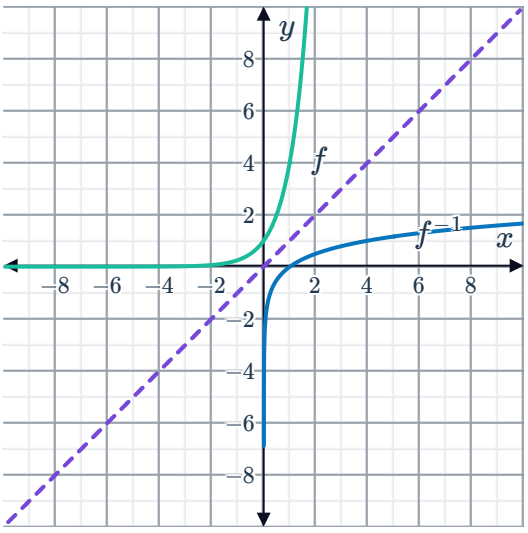
16 Consider the graph of $f(x) = 4^x$:



a State the domain of the function f.

b State the range of the function f.

The graph of f is reflected about the line $y = x$ to produce the graph of f^{-1} as shown:



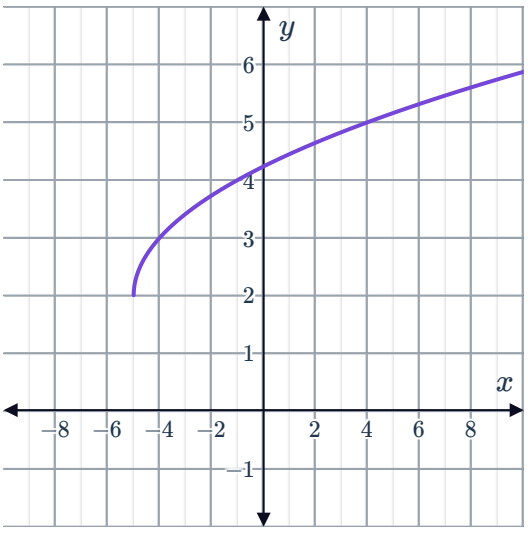
c State the domain of the function f^{-1}.

d State the range of the function f^{-1}.

e What is the domain of f equivalent to?

f What is the range of f equivalent to?

17 Consider the graph of $f(x) = \sqrt{x+5} + 2$:



a State the domain of its inverse function.

b State the range of its inverse function.

c Find the inverse function.

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18 Consider the function $f(x) = 0.8x + 0.8$.

- **a** Sketch a graph of $f(x)$ by determining any two points on the line.
- **b** Sketch a graph of the inverse f^{-1}.
- **c** State the domain of f.
- **d** State the range of f.
- **e** State the domain of f^{-1}.
- **f** State the range of f^{-1}.

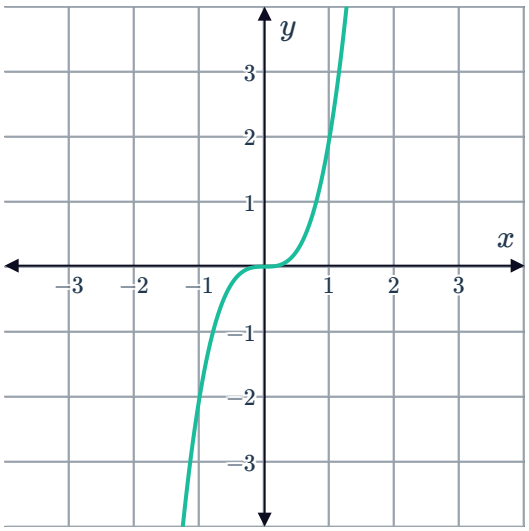
19 For each of the following functions:

- **i** Sketch the function $f(x)$ over its domain.
- **ii** Find the inverse, f^{-1}.
- **iii** State the domain of f^{-1}.
- **iii** State the range of f^{-1}.
- **iv** Sketch the function f^{-1} over its domain.

- **a** $f(x) = x + 6$ defined over the interval $[0, \infty)$.
- **b** $f(x) = 7 - x$ defined over the interval $[2, 9]$.
- **c** $f(x) = (x - 6)^2 - 2$ defined over the interval $[6, \infty)$.
- **d** $f(x) = \sqrt{4 - x}$ defined over the interval $[0, 4)$.
- **e** $f(x) = (x + 2)^2 + 3$ defined over the interval $[0, \infty)$.

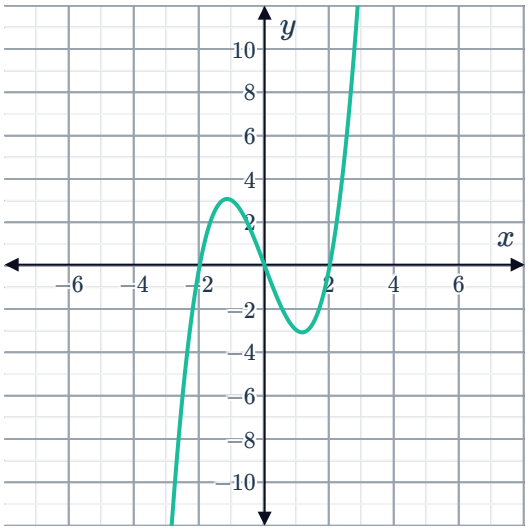
20 Consider the given graph:

- **a** Does this graph have an inverse function?
- **b** State the domain of this graph.
- **c** State the range of the inverse function of this graph.
- **d** Sketch the graph of the inverse function of this graph.



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21 Consider the given graph:



- **a** Does this graph have an inverse function?
- **b** State the domain of this graph.
- **c** State the domain restrictions that would allow the graph to have an inverse function.
- **d** If the domain for the original function is restricted to only one of the restrictions, what would be the range of the inverse function?

22 The tax on a new tablet is 7% of the advertized price, A.

- **a** Determine the equation for the total cost T as a function of the advertized price A.
- **b** Now, express the advertized price A as a function of the total cost T.

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